FINC5001

Module 4

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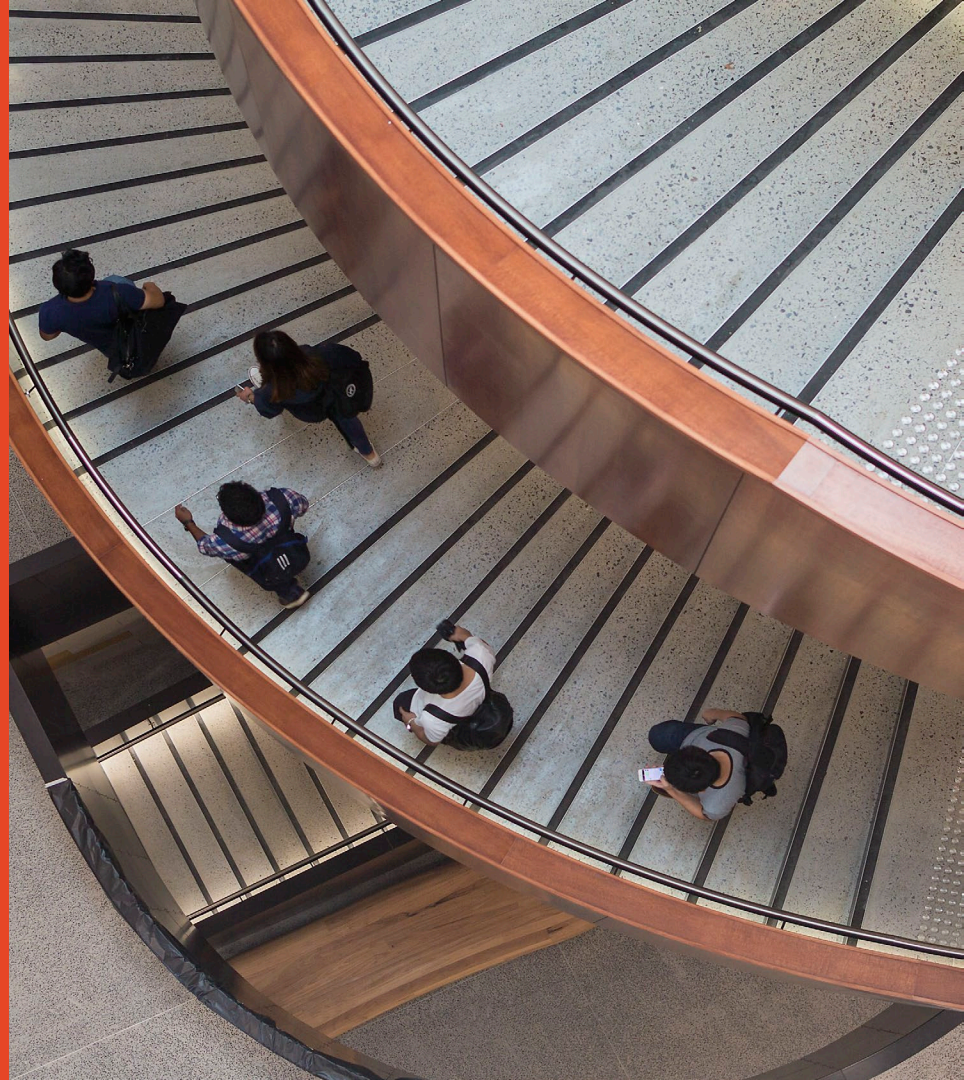


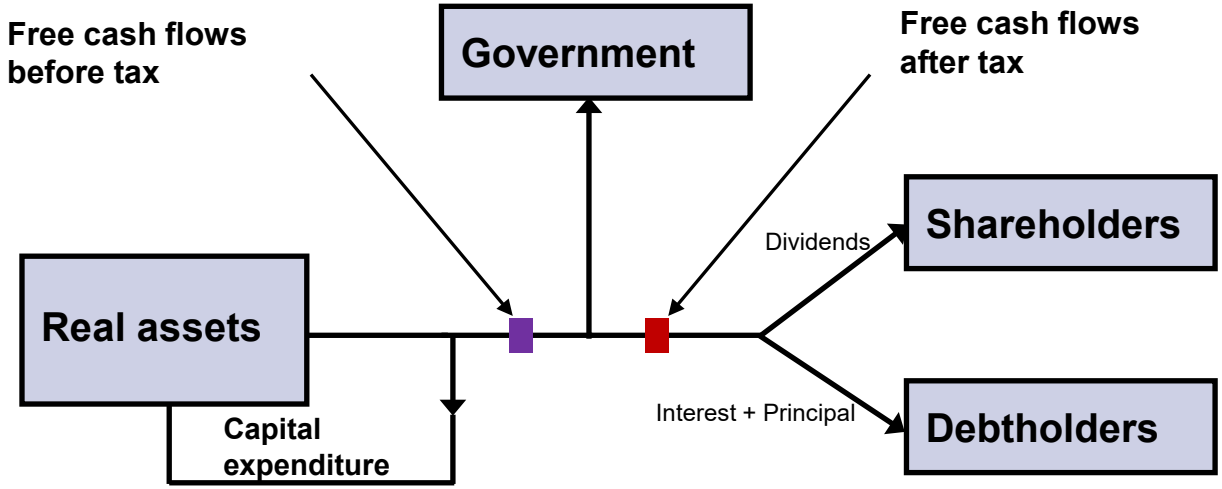
This Lecture

- What you should know by the end of this lecture
  - Application of financial mathematics
  - How to value a firm
  - How to value debt and equity securities
  - Share valuation models

Valuing a company

- The value of an asset is a function of:
  - The cash flows generated by the asset, *and*
  - The appropriate discount rate applied to these cash flows
- A firm can be valued by :
  - Calculating the present value of the cash flows generated by the firm's real assets
  - Calculating the present value of the firm's individual securities (i.e. equity and debt securities)

Valuing a company (cont)



- Cash flows generated by a firm's real assets ultimately flow through to the shareholders and debtholders of the firm as dividends and interest
- Hence the present value of cash flows generated must be equal to the value of cash flows received

Valuing a company (cont)

- The value of a company can be determined using the following equation:

$$V = D + E$$

where:

V = the present value of cash flows generated by the firm

D = the present value of cash flows generated by debt securities

E = the value of cash flows generated by equity securities

Value of a company's real asset cash flows

- The net cash flows from real assets, after reinvestment costs, are known as *free cash flows*
- If the firm is assumed to have an infinite life, the value of the firm is given by:

$$V = \sum_{t=1}^{\infty} \frac{F_t}{(1+r)^t}$$

where:

F_t = free cash flow generated by the company in period t

r = required rate of return (i.e. appropriate discount rate for the company)

Debt versus equity

- The alternative approach to valuing a firm involves calculating the present value of the individual debt and equity securities that make up the firm's *capital structure*, and adding them together
- There is a wide variety of debt, equity and hybrid securities currently available in capital markets
- We will focus on a simplified capital structure based on debt and equity only

Debt

In general, debt is a security which:

1. Pays a *contractually set* rate of return (interest)
2. Has an *finite* life
3. Debt security holders have a *first claim* to the cash flows of the company

Equity

In general, equity is a security which:

1. Pays a *uncertain* rate of return (dividends)
2. Has an *indefinite* life
3. Shareholders have a *residual claim* to the cash flows of the company

Characteristics of debt versus equity

Characteristic	Debt	Equity
Return	Fixed (interest)	Uncertain (dividend)
Life	Fixed (redeemed)	Indefinite
Security (claim)	First	Residual

Corporate borrowings

– There are two main sources of borrowing for a corporation:

1. Loans from a financial institution (known as **private debt** since the debt is not publicly traded)
   - » **Advantages** of private debt placement
     - speed
     - reduced costs
     - financing flexibility
   - » **Disadvantages** of private debt placement
     - interest costs
     - restrictive covenants
2. Bonds (known as **public debt** since they can be traded in the public financial markets)

Floating–rate loans

- In the private financial market, loans are typically **floating rate loans**, i.e. the interest rate is adjusted based on a specific benchmark rate. The most popular international benchmark rate is the **London Interbank Offered Rate (LIBOR)**, the rate at which banks offer to lend in the London wholesale or interbank market
- In Australia, a more common benchmark is the **bank bill swap rate (BBSW)**, which is the rate used for bank bill interest rate swaps, and therefore represents the interest rate at which investors are indifferent between a fixed or floating interest rate

Borrowing money in the public financial market

- Corporations engage the services of an **investment banker** while raising long-term funds in the public financial market. The investment banker performs three basic functions:
  - **Underwriting**: assuming risk of selling a security issued. The client is given the money before the securities are sold to the public.
  - **Distributing**
  - **Advising**

Corporate bonds

- **A corporate bond** is a debt security issued by a corporation that has promised future payments and a maturity date
- If the firm fails to pay the promised future payments of interest and principal, the bond trustee can classify the firm as insolvent and force the firm into bankruptcy
- The Australian bond market, while not as developed as that in other countries, is showing signs that it is starting to improve, with some large Australian companies issuing domestic bonds for the first time in recent years.

Basic bond features

- The basic features of a bond include the following:
  - bond indenture
  - claims on assets and income
  - face value (or par value)
  - coupon rate
  - maturity and repayment of principal
  - call provision and conversion features

Bond terminology

Indenture	The legal agreement between a firm issuing the bonds and the bond trustee who represents the bondholders. It lists the specific terms of the loan agreement, including a description of the bonds, the rights of the bondholders, the rights of the issuing firm and the responsibilities of the trustee.
Priority of claim on assets and income	In the case of insolvency, claims of debt in general, including bonds, are honoured before those of both ordinary and preference shares. In addition, interest payments hold priority over dividend payments for ordinary and preference shares.
Face value	The face value of a bond, also known as its par value, is the principal that must be repaid to the bondholder at maturity. Australian Treasury bonds are issued with face values of $1000. In some markets, such as the United States, when bond prices are quoted in the financial press, prices are expressed as a percentage of the bond's face value.
Maturity and repayment of principal	The maturity date refers to the date on which the bond issuer must repay the principal or face value to the bondholder.
Coupon rate	The coupon rate on a bond indicates the percentage of the face value of the bond that will be paid out annually in the form of interest.
Current yield	The current yield on a bond is the rate of return you will receive if you buy the bond at its current price and hold it until maturity. It is the discount rate that equates the present value of the future cash flows from the bond with its current price. (This will be covered more detail later in this chapter.)
Call provision	Some corporate bonds have a call provision, which provides the issuer of the bond with the right to redeem or retire a bond before it matures.
Conversion feature	In addition, some corporate bonds have a conversion feature that allows bondholders to convert their bonds into a set number of ordinary shares.

Public Debt Valuation

- **Key Concept**
  - Value of debt is the present value of future cashflows stemming from the debt
- Cashflows stemming from debt:
  - Interest (C) = 'coupon' which is the dollar amount of interest paid periodically
  - Face value (B) = lump sum paid at end of life of debt security

Valuation of corporate debt

- This would result in a bond formula that looks like this:

$$D = C\left[\frac{1-(1+r_d)^{-n}}{r_d}\right] + \frac{F}{(1+r_d)^n}$$

- Where:
- C = coupon payment made by the bond each period
- F = face value of the bond
- r_d = required return on debt (yield to maturity)
- n = number of periods until the bond reaches maturity

Valuation of corporate debt (cont)

Example

On 23 November 2009, Brookfield Multiplex Ltd listed secured bonds carrying a coupon of 8.79% p.a., payable quarterly, at a face value of \$100 and maturing on 23 November 2012. Assuming that the notes were issued at a yield of 8.10% p.a., what was the value of these securities?

Interest payment = coupon rate × face value

$= 0.0879/4 \times \$100 = \$2.20$ per quarter

There are 12 quarterly payments

Yield = 8.10% p.a., so the effective quarterly yield is 2.0%

Valuation of corporate debt (cont)

Example

On 23 November 2009, Brookfield Multiplex Ltd listed secured bonds carrying a coupon of 8.79% p.a., payable quarterly, at a face value of $100 and maturing on 23 November 2012. Assuming that the notes were issued at a yield of 8.10% p.a., what was the value of these securities?

$$D = C\left[\frac{1-(1+r_d)^{-n}}{r_d}\right] + \frac{F}{(1+r_d)^n}$$

$$= 2.20\left[\frac{1-(1.02)^{-12}}{0.02}\right] + \frac{100}{1.02^{12}} = \$102.12$$

Note - A Convention

- Dollar coupon (C) = coupon rate x face value
- Coupon rate is placed in front of name of security
  - eg. 3.5% p.a. December 2025 Commonwealth Bonds
- Why isn't the day listed?
  - All Commonwealth bonds make interest payments and mature on the 15th of the month
- Yield = discount rate in debt markets i.e. $\mathbf{r_d}$
- **Commonwealth Bond coupon payments and yields are compounded semi-annually (even if not explicitly stated)**

Bond ratings and default risk

- **Bond ratings** indicate the default risk, i.e. the probability that the firm will make the bond's promised payments. Rating agencies use borrower's financial statements, financing mix, profitability, variability of past profits and make judgments about the quality of the firm's management in order to determine ratings

Interpreting bond ratings

Bond rating category	Standard & Poor's	Moody's	Description
Investment grade:			
Prime or highest strong	AAA	Aaa	Highest quality; extremely strong capacity to pay
High quality	AA	Aa	Very strong capacity to pay
Upper medium	A	A-1, A	Upper medium quality; strong capacity to pay
Medium	BBB	Baa-1, Baa	Lower medium quality; changing circumstances could impact the firm's ability to pay
Not investment grade:			
Speculative	BB	Ba	Speculative elements; faces uncertainties
Highly speculative	B, CCC, CC	B, Caa, Ca	Extremely speculative and highly vulnerable to non-payment
Default	D	C	In default, with little prospect of recovery of principal or interest

Using market-yield-to-maturity data

- Market-yield-to-maturity data are regularly reported by a number of investor services and are quoted in terms of **credit spreads** or **spreads to Treasury bonds**. Table 9.4 contains some examples of yield spreads.

Corporate bond spread tables

Investment Grade

Junk Bonds

	Maturity						
Rating	1 yr	2 yr	3 yr	5 yr	7 yr	10 yr	30 yr
Aaa/AAA	5	8	12	18	28	42	65
Aa1/AA+	10	18	25	34	42	54	77
Aa2/AA	14	29	38	50	57	65	89
Aa3/AA−	19	34	43	54	61	69	92
A1/A+	23	39	47	58	65	72	95
A2/A	24	39	49	61	69	77	103
A3/A2−	32	49	59	72	80	89	117
Baa1/BBB+	38	61	75	92	103	115	151
Baa2/BBB	47	75	89	107	119	132	170
Baa3/BBB−	83	108	122	140	152	165	204
Ba1/BB+	157	182	198	217	232	248	286
Ba2/BB	231	256	274	295	312	330	367
Ba3/BB−	305	330	350	372	392	413	449
B1/B+	378	404	426	450	472	495	530
B2/B	452	478	502	527	552	578	612
B3/B−	526	552	578	604	632	660	693
Caa/CCC+	600	626	653	682	712	743	775
US Treasury yield	0.13%	0.45%	0.93%	1.74%	2.31%	2.73%	3.55%

Source: Bonds Online, <www.bondsonline.com/Todays_Market/Corporate_Bond_Spreads.php>.

$YTM_{30\text{ yr. Aa2/AA}}$
$= \text{Spread} + YTM_{\text{Treasury bond of like maturity}}$
$= 0.89\% + 3.55\% = 4.44\%$

$YTM_{30\text{ yr. Ba1/BB+}}$
$= \text{Spread} + YTM_{\text{Treasury bond of like maturity}}$
$= 2.86\% + 3.55\% = 6.41\%$

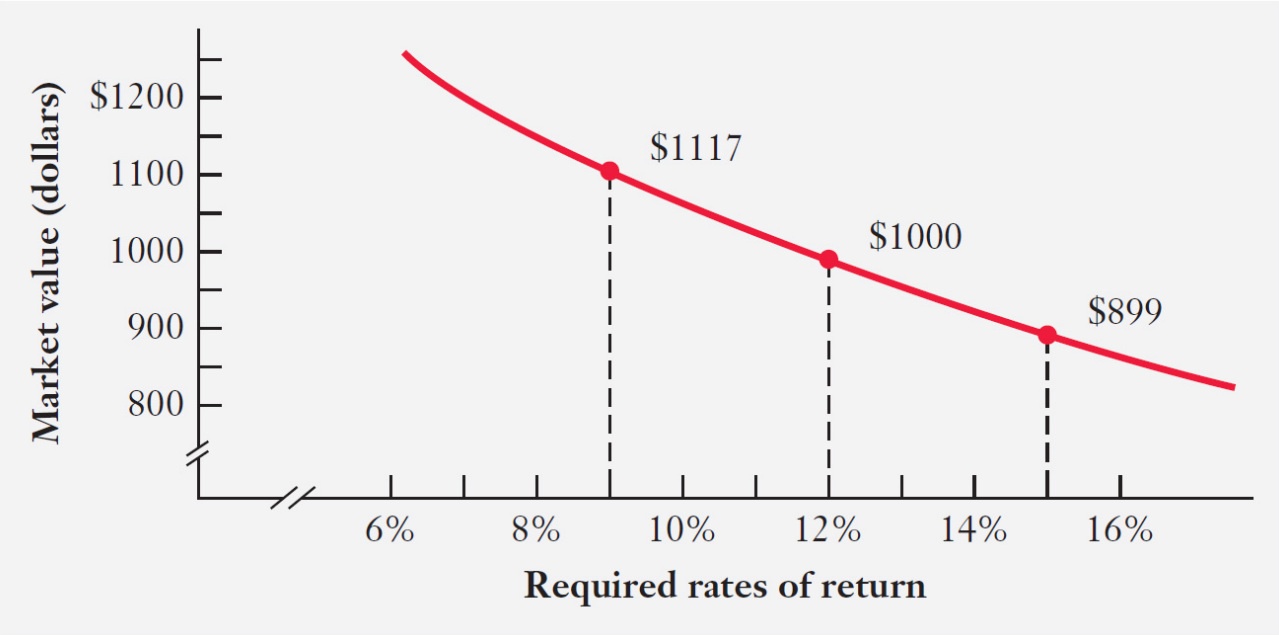
Bond valuation: Four key relationships

– First relationship: The value of a bond is inversely related to changes in the market's required yield to maturity.

	YTM = 12%	YTM rises to 15%
Par value	$1000	$1000
Coupon rate	12%	12%
Maturity date	5 years	5 years
Bond value	$1000	$899.44

Bond Value Drops

Bond value and the market's required yield to maturity



Bond valuation: Four key relationships (cont.)

- Second relationship: The market value of a bond will be less than the face value (**discount bond**) if the market's required yield to maturity is above the coupon interest rate and will be greater than face value (**premium bond**) if the market's required yield to maturity is below the coupon interest rate.

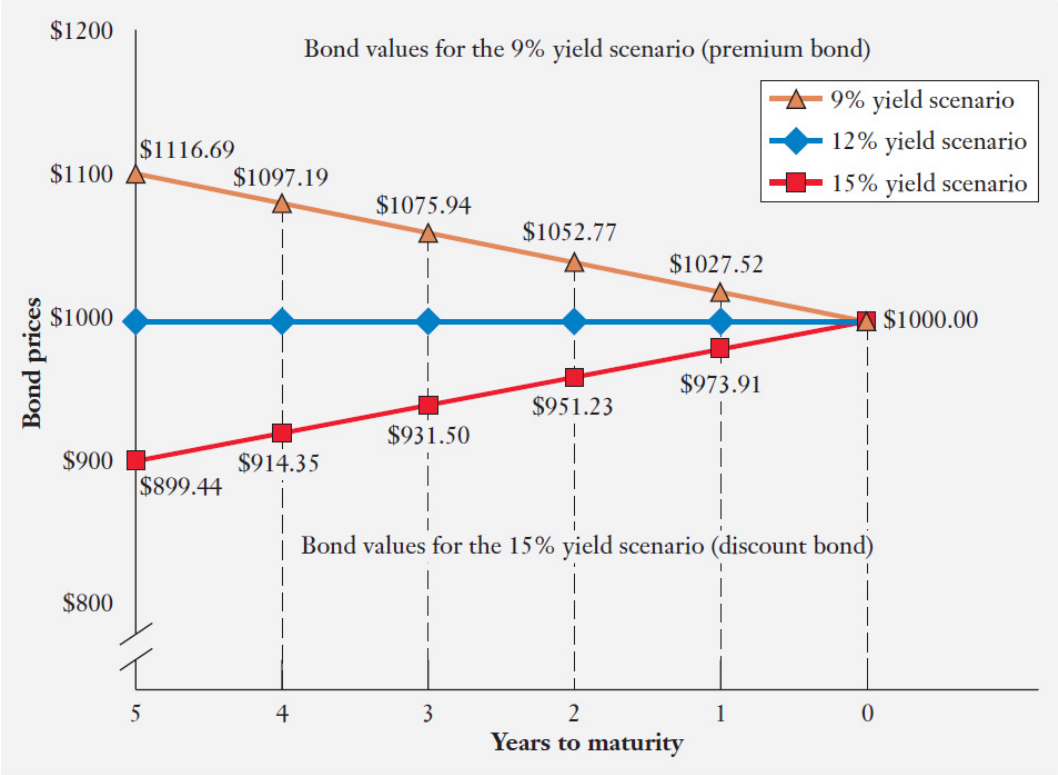
Bond valuation: Four key relationships (cont.)

- Third relationship: As the maturity date approaches, the market value of a bond approaches its face value. This is because at maturity the bond will be taken away and the investor will receive the face value of the bond.

Bond prices relative to maturity date

		Years to maturity					
	12% coupon bond	**5**	**4**	**3**	**2**	**1**	**0**
	12% yield scenario	$1000.00	$1000.00	$1000.00	$1000.00	$1000.00	$1000.00
Discount bond	15% yield scenario	$ 899.44	$ 914.35	$ 931.50	$ 951.23	$ 973.91	$1000.00
Premium bond	9% yield scenario	$1116.69	$1097.19	$1075.94	$1052.77	$1027.52	$1000.00

Value of a 12% coupon bond during the life of the bond



Bond valuation: Four key relationships (cont.)

- Fourth relationship: Long-term bonds have greater interest-rate risk than short-term bonds.

- While all bonds are affected by a change in interest rates, the prices of longer-term bonds fluctuate more when interest rates change than do the prices of shorter-term bonds.

Bond valuation: Four key relationships (cont.)

	Years to maturity					
	5	10	15	20	25	30
15% (Increased yield)	$ 899.44	$ 849.44	$824.58	$ 812.22	$ 806.08	$ 803.02
% price decrease	−10.1%	−15.1%	−17.5%	−18.8%	−19.4%	−19.7%
12% (Base case)	**$1000.00**	**$1000.00**	**$1000.00**	**$1000.00**	**$1000.00**	**$1000.00**
% price increase	11.7%	19.3%	24.2%	27.4%	29.5%	30.8%
9% (Decreased yield)	$1116.69	$1192.53	$1241.82	$1273.86	$1294.68	$1308.21

Valuation of ordinary shares

- Equity securities are valued by calculating the present value of all cash flows generated
- These cash flows consist of:

  1. Dividends
  2. The share price

- The current value of a share is, in part, influenced by its future value.
- However, the future share price is also based on the value of future dividends

Valuation of ordinary shares (cont)

– Consequently, the value of a share is given by the present value of all future dividends that the company is expected to pay:

$$E = \sum_{t=1}^{\infty} \frac{d_t}{(1 + r_e)^t}$$

where:

E = the value of equity

d_t = the dividend paid in period t

r_e = the cost of the equity

Dividend valuation models

- Constant-dividend model
  - Assumes that companies pay a constant dividend
  - The value of a share can be expressed:

$$P_o = \frac{d}{r_e}$$

where:

P_0	=	current value of share
d	=	(constant) dividend per period
r_e	=	required return on the share

Constant dividend model

Example

The cash dividend by Telstra each six months has been 20 cents per share (including franking credits). If investors are expecting a return of 4.5% each six months, what would be Telstra's share price on 19 February 2013?

$$P_0 = \frac{d}{r_e} \therefore P_0 = \frac{0.20}{0.045}$$

$$\therefore P_0 = \$4.44$$

Telstra Dividend History

Previous Dividend Payments

Ex Date	Amount	Franked %	Books Closed	Payable
01 Mar 2016	0.1550	100	03 Mar 2016	01 Apr 2016
25 Aug 2015	0.1550	100	27 Aug 2015	25 Sep 2015
25 Feb 2015	0.1500	100	27 Feb 2015	27 Mar 2015
27 Aug 2014	0.1500	100	29 Aug 2014	26 Sep 2014
24 Feb 2014	0.1450	100	28 Feb 2014	28 Mar 2014
19 Aug 2013	0.1400	100	23 Aug 2013	20 Sep 2013
18 Feb 2013	0.1400	100	22 Feb 2013	22 Mar 2013
20 Aug 2012	0.1400	100	24 Aug 2012	21 Sep 2012
20 Feb 2012	0.1400	100	24 Feb 2012	23 Mar 2012
22 Aug 2011	0.1400	100	26 Aug 2011	23 Sep 2011
21 Feb 2011	0.1400	100	25 Feb 2011	25 Mar 2011
23 Aug 2010	0.1400	100	27 Aug 2010	24 Sep 2010
22 Feb 2010	0.1400	100	26 Feb 2010	26 Mar 2010
24 Aug 2009	0.1400	100	28 Aug 2009	25 Sep 2009
06 Mar 2009	0.1400	100	13 Mar 2009	09 Apr 2009
25 Aug 2008	0.1400	100	29 Aug 2008	26 Sep 2008
03 Mar 2008	0.1400	100	07 Mar 2008	04 Apr 2008
20 Aug 2007	0.1400	100	24 Aug 2007	21 Sep 2007
26 Feb 2007	0.1400	100	02 Mar 2007	30 Mar 2007
21 Aug 2006	0.1400	100	25 Aug 2006	22 Sep 2006

Source: SMH quotes

Dividend valuation models (cont)

- Constant-growth model
  - Assumes that dividends grow at a constant rate
  - The value of a share can be expressed:

$$P_0 = \frac{d_1}{r_e - g}$$

where:

d_1	=	$d_0(1+g)$ = dividend at end of the first period
r_e	=	the equity cost of capital
g	=	the constant growth rate (g must be < r_e)

Constant dividend growth model

Example

Shareholders in XYZ Ltd received a semi-annual dividend of $0.10 today. The dividend is expected to grow at a rate of 1.5% every six months. The shareholders of XYZ require a return on 15%pa compounded annually on their investment. What should be the market price of XYZ one share today?

Effective required rate of return = $1.15^{0.5}$ -1 = 0.0724

$$P_0 = \frac{d_0(1+g)}{r_e - g}$$

$$P_0 = \frac{0.1(1+0.015)}{0.0724 - 0.015} = \$1.77$$

Estimating growth rate, g

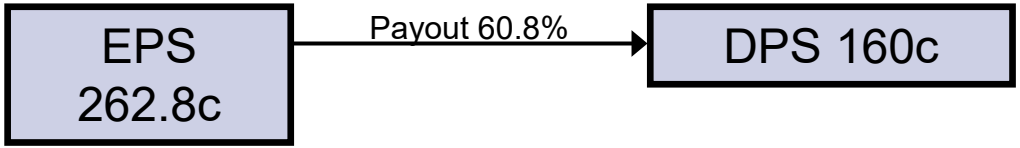
- Estimating a company's growth rate, *g*, is a key issue in the application of the perpetual growth model
- According to this model, dividend growth is derived from the earnings of a company
- *Case Study- ANZ Bank*
- For the year ended September 2018:
  - Earnings per share (EPS) = 262.8 cents
  - Dividends per share (DPS) = 160 cents
  - Return on (book) equity (ROE) = 10.3%

Estimating growth rate (cont)

EPS
262.8c

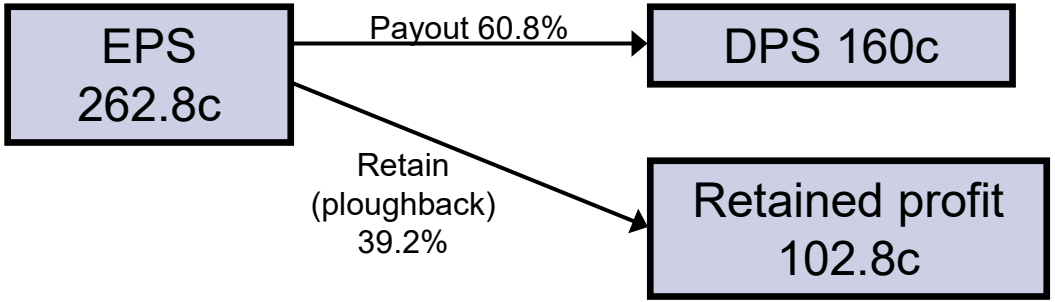
- The EPS represents the profit generated for each share – management must decide whether to pay this out as dividends and/or retain it in the firm

Estimating growth rate (cont)



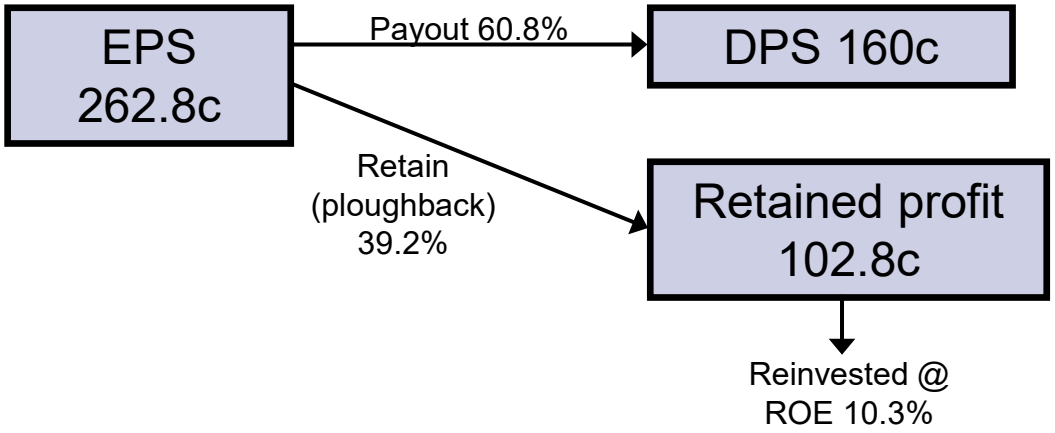
- In 2018, ANZ paid out 160c (60.8% of its EPS) as a dividend
- The proportion of EPS paid out as dividends is called the ‘payout ratio’

Estimating growth rate (cont)



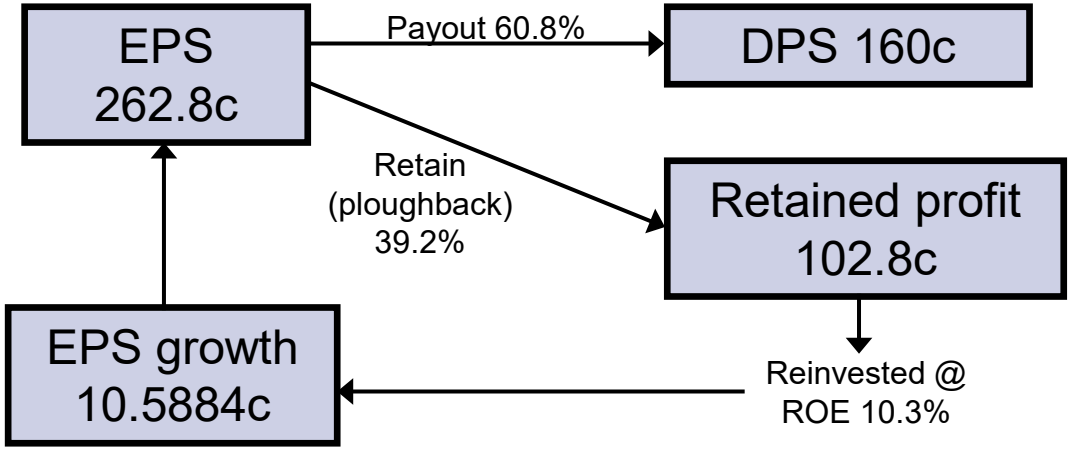
- The remaining 102.8c or 39.2% of the EPS is retained within the firm to finance future growth
- This is sometimes called the ‘ploughback ratio’

Estimating growth rate (cont)



- The 39.2% in retained profit is reinvested within the firm, and it is assumed that it will generate earnings at the firm's current ROE of 10.3%

Estimating growth rate (cont)



- This yields 10.5884c in increased EPS in the future, which will in turn be available for distribution as dividends or ploughed back into the firm

Estimating growth rate (cont)

- Using this model, the dividend growth rate can be expressed as follows:

$$g = (1 - \text{payout ratio}) \times ROE$$

- Assumptions:
  - The payout ratio is stable
  - The estimated ROE is stable

Estimating growth rate (cont)

Example

In the year ended 30 September 2018, ANZ Banking Group (ANZ) generated EPS of 262.8 cents, paid a dividend of 160 cents per share, and its ROE was estimated at 10.3. What is the expected future earnings and dividend growth rate?

$$Payout\ ratio = 160/262.8 = 0.608$$

$$g = (1 - \text{payout ratio}) \times ROE$$

$$= (1 - 0.608) \times 0.103 = 0.040376 = 4.0376\%$$

Features of preference shares

- **Dividend**: In general, size of preference shares dividend is *fixed*, and it is either stated as a dollar amount or as a percentage of the preference share's par value
- **Multiple classes**: A company can issue more than one class of preference shares, and each class can have different characteristics
- **Claims on assets and income:** Preference shares have priority over those of ordinary shareholders for payment of dividends and in settlement of claims at bankruptcy
- Most preference shares carry a **cumulative feature** (i.e. all past unpaid dividends must be paid before any ordinary shares dividends can be declared)

Features of preference shares (cont.)

	Symbol	Face value	Issue date	Conversion date*	Margin	Price	Yield
Convertible Preference Shares 2	ANZPA	$100.00	17/12/2009	15/12/2016	3.10% p.a.	$100.200	4.0%
Convertible Preference Shares 3	ANZPC	$100.00	28/09/2011	1/09/2019	3.10% p.a.	$99.000	4.0%
Capital Notes 1	ANZPD	$100.00	28/09/2011	1/09/2023	3.40% p.a.	$99.100	4.3%
Capital Notes 2	ANZPE	$100.00	28/09/2011	24/03/2024	3.25% p.a.	$98.400	4.2%

* Mandatory conversion to ordinary shares occurs on this date, subject to certain conditions being met, or after this date once those conditions are met. Subject to other conditions, the issuer may redeem or convert prior to this date.

Sources: Australian Securities Exchange <www.asx.com.au/asx/markets/interestRateSecurityPrices.do?type=HYBRID>; ANZ CP2 Prospectus <http://phx.corporate-ir.net/phoenix.zhtml?c=96910&p=irol-prospectusdisclaimer>; ANZ CP3 Prospectus <www.fundsfocus.com.au/managed-funds/pdfs/ANZ/anzpc-prospectus.pdf>; ANZ CN1 Prospectus <http://privatewealth.nab.com.au/resources/NabPrivate/pdf/anz_capital_offer_prospectus.pdf>; and ANZ CN2 Prospectus <www.asx.com.au/documents/products/ANZ_Capital_Notes_2_Prospectus_11_Feb_14.pdf>.

Valuing preference shares

- Preference share valuation
  - Preference shares pay a fixed dividend yield
  - The value of a preference share can be expressed:

$$P_o = \frac{d}{r_{ps}}$$

where:

P_0	=	current value of share
d	=	(constant) dividend per period
r_{ps}	=	required return on the preference share

Estimating the market's required yield

- Estimating the **market yield**:
  - Can be found by re-arranging the previous equation

$$r_{ps} = \frac{d}{P_0}$$

where:

P_0	=	current value of share
d	=	(constant) dividend per period
r_{ps}	=	required return on the preference share

The End

Next week we begin to examine Risk-adjusted Discounted Cash Flows, Investor Preferences, Diversification and Portfolios